ONETECH

Accredited Testing Laboratory under the terms of ISO/IEC 17025

ENVIRONMENTAL TEST REPORT

for

Product: Transformer

"MODEL NAME: TRF"

APPLICANT

Woojin Industrial Systems Co., Ltd.

613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, Korea

October 31, 2012

Test Results

"Refer to the test results attached"

1. As a test result of sample which was submitted from the client, this test report does not guarantee the whole product quality.

2. This test report should not be used for public relations, advertisement, conduct propaganda and raised for a suit without a written agreement of testing institute, and should not be used for various purpose beside subscribed use.

3. This test report including test result conducted by using the test method which was suggested form a client.

G. W. Lee **(signature)**

Goan Woo Lee / Managing Director

***This is only valid in connection with Test Report:* R12OR-039**

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TEST REPORT

Report No .....:	R12OR-039	
Date of issue .....:	October 31, 2012	
Testing Laboratory .....:	ONETECH Corp.	AGR. No.: A12OA-168
Address .....:	301-14 Daessangnyeong-ri, Chowol-eup, Gwangju-si, Gyeonggi-do 464-862 Korea	
Test location .....:	Same as above	
Applicant	Woojin Industrial Systems Co., Ltd.	
Address .....:	613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, Korea	
Standard .....:	Test specification supplied by the manufacturer, (Document No. 423-316, Section No. 7.3)	
Purpose of use .....:	Quality Control	
Number of pages (Report).....:	9 Pages	
Number of pages (Attachments) .....:	N/A	

Prepared by ..... :		Reviewed by.....:	
(+ signature)	Dong Kyun Park / Engineer	(+ signature)	Jin Won Seo / Project Engineer

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1. CLIENT AND MANUFACTURER INFORMATION

-. Client : Woojin Industrial Systems Co., Ltd.

-. Address : 613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, KOREA

-. Manufacturer : Woojin Industrial Systems Co., Ltd.

-. Address : 613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, KOREA

-. Telephone No. : +82-43-820-4263

-. Fax No. : +82-43-836-7653

2. TEST SUMMARY

2.1 Test items and results

STANDARDS		RESULTS
Test specification supplied by the manufacturer (Document No. 423-316, Section No. 7.3)	Damp Heat Test	PASS

2.2 Criterion description

Measurement	
Initial measurement	It checks normal operating of the sample before test starts.
Intermediate measurement	After the first cyclic is finished, it checks normal operating of the sample with enough dry status.
Final measurement	It checks normal operating of the sample with enough dry status at the end of test.

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2.3 Basis test condition

Items		Descriptions
Initial Measurements		- No crack, transformation and swell should be visually not found. - After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT should be no differences in functions of operation stated in the section 3.3.
Precondi-tioning	Regulations	The regulations of preconditioning is not required : ■ The regulations of preconditioning is required : □
	Contents	N/A
Intermediate Measurements	Regulations	The regulation of intermediate measurements is not required : □ The regulation of intermediate measurements is required : ■
	Contents	- After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT should be no differences in functions of operation stated in the section 3.3.
Recovery	Regulations	The regulation of recovery is not required : ■ The regulation of recovery is required : □
	Contents	N/A
Final Measurements		- No crack, transformation and swell should be visually not found. - After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT should be no differences in functions of operation stated in the section 3.3.

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3. EUT (Equipment Under Test)

3.1 Identification of EUT

-. Equipment : Transformer

-. Type/model : TRF

-. Brand Name : N/A

-. Serial number : N/A

-. Manufacturer : Woojin Industrial Systems Co., Ltd.

3.2 Configuration

Model	Manufacturer	Description	Connected to
TRF	Woojin Industrial Systems Co., Ltd.	Transformer (EUT)	-
CRA	Woojin Industrial Systems Co., Ltd.	Communication Rack A	EUT

3.3 Operational mode of EUT

- After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT maintains normal operations.

3.4 EUT modifications

- N/A

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4. DAMP HEAT TEST

The **Damp Heat Test** was performed at ONETECH Corp. **Date** : October 19 ~ October 21, 2012

4.1 Environmental condition

Ambient temperature	22 ℃
Relative humidity	50 % R.H.
Air pressure	101 kPa

4.2 Test equipment used

	Model Number	Manufacturer	Description	Serial Number
■	ER-55MHHP	HITACHI	TEMPERATURE AND HUMIDITY CHAMBER	U5972178

4.3 Test condition

4.3.1 Detailed test condition

Items		Descriptions
Standard		Test specification supplied by the manufacturer
Operational mode of EUT		Non-operational test : ■ Operational test : □
Method for change of temperature		Rapid change of temperature : □ Regular change of temperature : ■
Status of heat -dissipating		Heat-dissipating : □ Non heat-dissipating : ■
Initial Measurements		There is same as the initial measurements stated in the section 2.3.
Precondi-tioning	Regulations	The regulations of preconditioning is not required : ■ The regulations of preconditioning is required : □
	Contents	N/A
Intermediate Measurements	Regulations	The regulation of intermediate measurements is not required : □ The regulation of intermediate measurements is required : ■
	Contents	See the next graph
Recovery	Regulations	The regulations of recovery is not required : ■ The regulations of recovery is required : □
	Contents	N/A
Final Measurements		There is same as the final measurements stated in the section 2.3.

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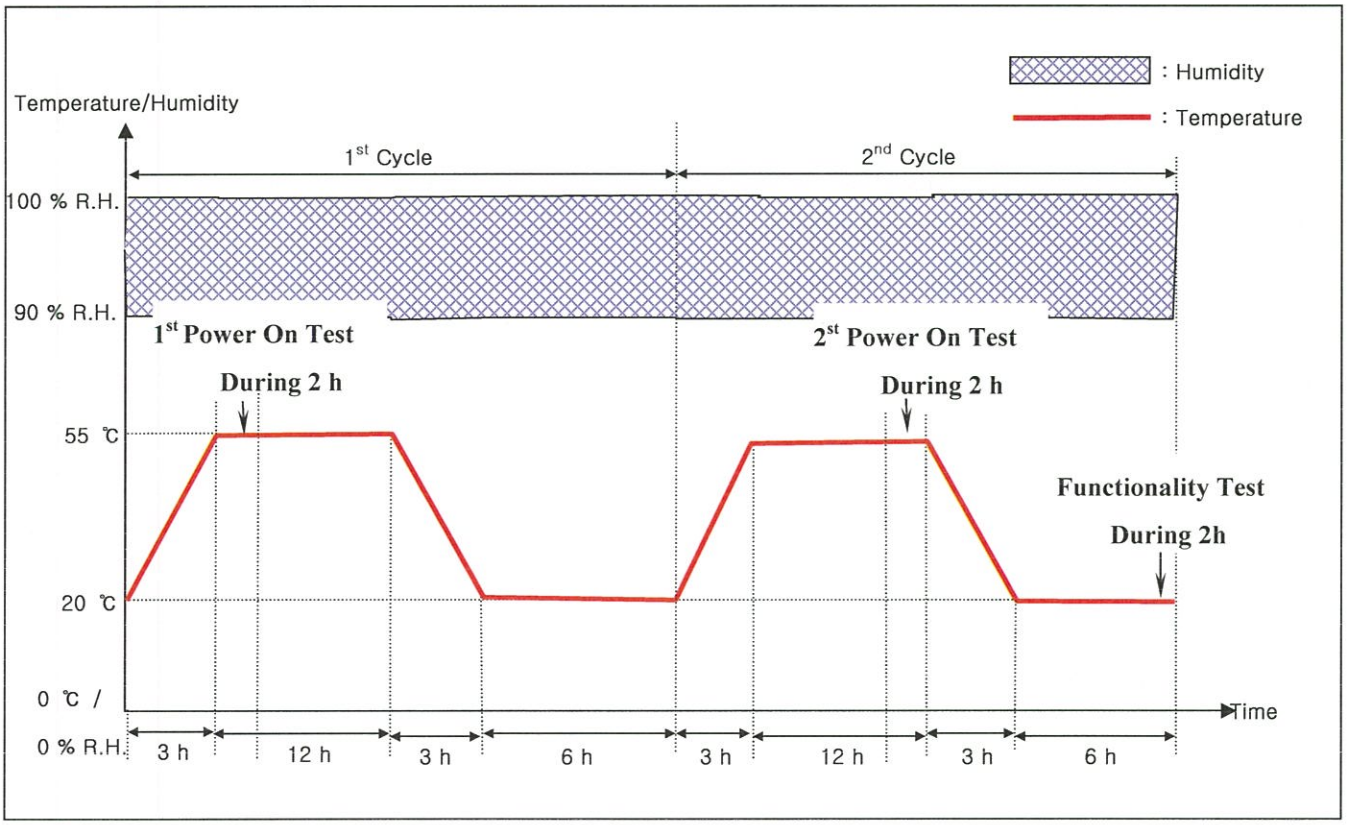
4.3.2 Test severities

Temperature			Humidity			Cycle Number	Test sequence
Set-up		Rate of change	Set-up		Rate of change		
Temp-erature	Time		Hum-idity	Time			
20 ℃ 55 ℃	12 h 24 h	See the below graph	95 % R.H.	48 h	See the below graph	2	[55 ℃(95 % R.H.) → 20 ℃(95 % R.H.)] = 2 Cycle

<Remark : Damp Heat Test Graph>



4.4 Test result

According to criterion description stated in the section 2.3, the initial measurements, intermediate measurements and final measurements were measured as follows.

EUT	Initial Measurements (1st Power On Test)	Intermediate Measurements (2st Power On Test)	Final Measurements (Functionality Test)
Sample 1	PASS	PASS	PASS

Tested by: Dong-Kyun Park / Engineer

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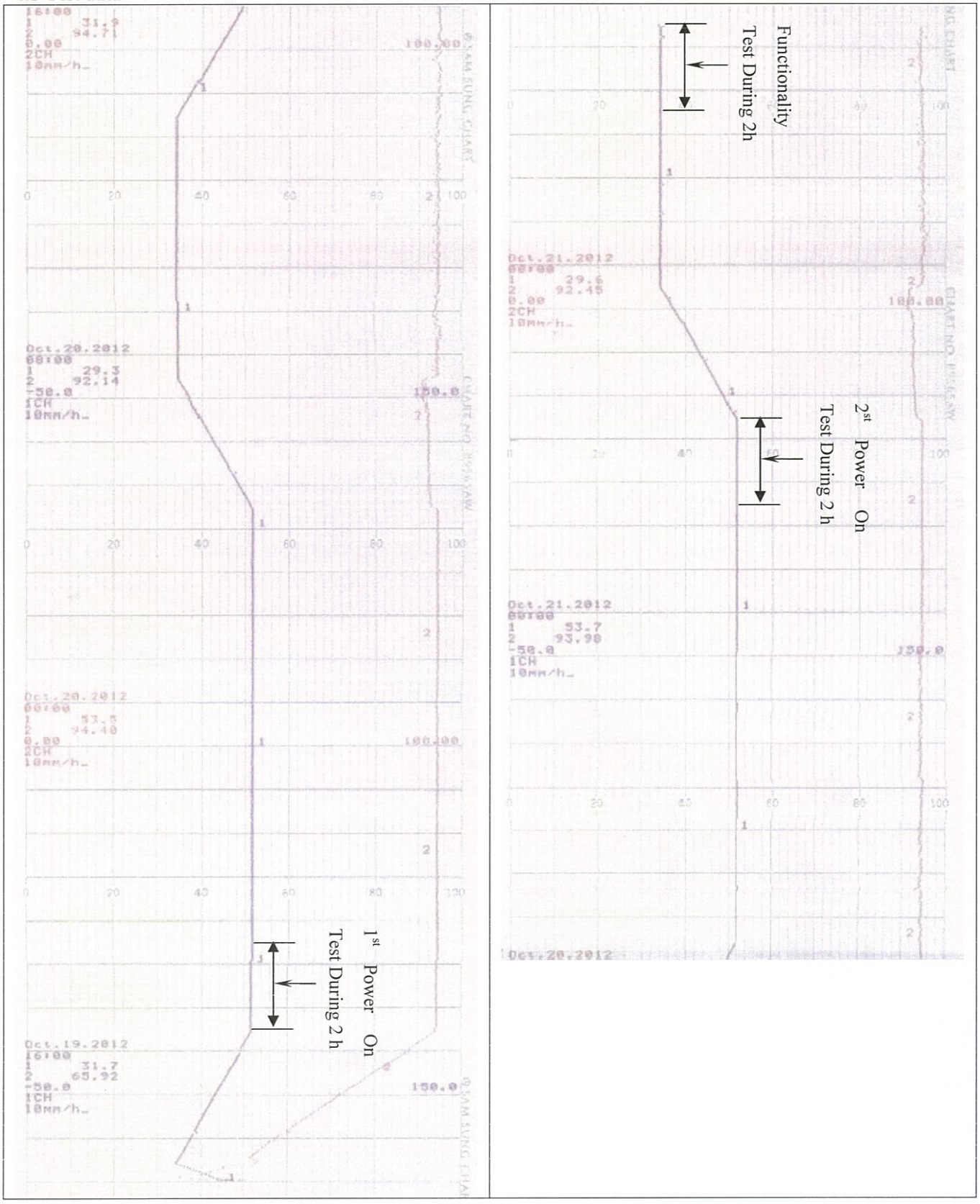
4.5 Test data


Functionality
Test During 2h
2st Power On
Test During 2 h
1st Power On
Test During 2 h

APPENDIX I - PHOTO OF CHAMBER TEST

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